llmXive Automated Review of OrbitQuant: Data-Agnostic Quantization for Image and Video Diffusion Transformers

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a strong case for OrbitQuant's effectiveness, with most claims supported by the provided tables and figures. However, there are minor instances of overgeneralization or ambiguity in the text that could mislead a reader about the scope of the results.

Specifically, the claim that OrbitQuant "exceeds FP16" is technically true for two models but not the third (FLUX.1-dev), and the text does not explicitly qualify this. Similarly, the statement that OrbitQuant is the "strongest PTQ method" on video backbones is true for the Overall Consistency metric but not for all individual metrics (e.g., Imaging Quality on CogVideoX-2B). The qualitative claim about baselines "collapsing to noise" on Z-Image-Turbo is supported by the SVDQuant data but could be clearer about which specific baselines were tested at that bit-width.

These are not fatal errors, as the data in the tables generally supports the authors' conclusions, but the text could be more precise to avoid implying universal dominance where it is metric-specific or model-specific. The ablation study claims are well-supported by the data, though the margin of improvement over Full RHT is small and could be described more precisely.

Overall, the factual claims are largely accurate, but a few sentences require qualification to ensure the reader understands the exact conditions under which the results hold.

Action items.

- [**writing**] The paper presents a strong case for OrbitQuant's effectiveness, with most claims supported by the provided tables and figures. However, there are minor instances of overgeneralization or ambiguity in the text that could mislead a reader about the scope of the results. Specifically, the claim that OrbitQuant "exceeds FP16" is technically true for two models but not the third (FLUX.1-dev), and the text does not explicitly qualify this. Similarly, the statement that OrbitQuant is the "strongest PT

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively communicates the high-level workflow of OrbitQuant, but the 3D plots lack axis labels for verification, and the mathematical notation in the central panel is slightly inconsistent with the caption's description of the rotation cancellation.

Figure 2

Figure 2 effectively visualizes the distribution of activation tokens across different rotation methods (Raw, Haar, RPBH) for two projection dimensions. The caption clearly defines all visual elements, including the dashed target curve, the KS distance insets, and the red codebook bin edges, ensuring the figure is self-contained and easy to interpret.

Figure 3

Figure 3 effectively presents a qualitative comparison of image and video generation across different quantization methods. The layout is clear, with distinct rows for models and columns for methods, and the captions provide sufficient context to interpret the visual results without ambiguity.

Figure 4

The figure effectively visualizes the trade-off between latency and memory, but the right panel is incomplete as it omits the QuaRot data point defined in the legend. Additionally, the latency units on the x-axis appear potentially incorrect or ambiguous, which hinders accurate interpretation of the performance gains.

Figure 5

The figure effectively communicates the ablation results, but the left panel's x-axis labels conflict with the caption's description of fixed activation precision, and the right panel's axis label lacks specificity regarding the compression scope.

Action items.

- [**science**] Figure 1: The mathematical notation in the central panel (\$W' ilde{x}' pprox (W i_d^ op)(i_d x) = Wx\$) is missing the rotation matrix i_d in the final equality term shown in the caption ($W'x'Wx$), creating a visual disconnect between the diagram's logic and the caption's claim that the rotation 'cancels'.
- [**writing**] Figure 1: The 3D surface plots in Panel 1 lack visible axis tick labels and units, making it impossible to verify the magnitude of the 'drift' described in the caption.
- [**science**] Figure 4: The legend defines four methods (OrbitQuant, SmoothQuant, QuaRot, ViDiT-Q), but the right panel (Video) only displays three data points (OrbitQuant, SmoothQuant, ViDiT-Q). The QuaRot data point is missing, preventing a complete comparison for the video task.

- [**writing**] Figure 4: The x-axis label 'Latency (s/img)' for the left panel implies a per-image metric, but the x-axis values (25-35) are unusually high for a single image inference step on modern hardware, suggesting the unit might be 'ms/img' or the metric is 'time per batch'. This ambiguity makes the performance claims difficult to verify.
- [**science**] Figure 5: The left panel's x-axis labels ('BF16', 'W4', 'W3', 'W2') contradict the caption's claim that 'AdaLN activations in BF16' are fixed; the labels imply the activations are being quantized alongside the weights.
- [**writing**] Figure 5: The right panel's y-axis label 'Compression ratio' is ambiguous; the caption clarifies it is 'model compression', but the axis label alone does not specify if this is total model size or just the AdaLN component.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written for a specialized audience, but it relies on several acronyms and notation conventions that are introduced without explicit definition, creating minor barriers for a competent reader from an adjacent field (e.g., a computer vision researcher not specializing in quantization or randomized linear algebra).

The primary issue is the premature use of the acronym **RPBH** (Randomized Permuted Block-Hadamard). It appears in the Abstract, Introduction, and Methodology sections before the full phrase is ever spelled out. While the full phrase appears in the caption of Figure 2, the acronym is used in the main text (e.g., Section 1, paragraph 3; Section 2.1) without a prior expansion. A reader skimming the text would encounter "RPBH" without knowing what it stands for until much later or by guessing.

Additionally, the notation **W4A4** (and similar variants like W2A4) is used extensively in the Results section and tables without a definition. While standard in the quantization subfield, an adjacent-field PhD might not immediately parse "W" as weight bits and "A" as activation bits. A brief parenthetical expansion at the first occurrence in Section 3.1 would resolve this.

The symbol **h** is introduced in Section 2.3 as the block size for the Hadamard transform but is not defined in the Notation subsection (Section 1.1) or immediately before Equation 5. While the text eventually explains it, the lack of a formal definition at the point of introduction forces the reader to infer the meaning from context.

Finally, terms like **Full RHT** and **Block-RHT** are used in the Ablation section (Section 4) without being defined as acronyms for "Randomized Hadamard Transform." Defining these at first use would ensure clarity.

These are minor, easily fixable issues that do not detract from the scientific contribution but would improve the self-containment of the paper for a broader technical audience.

Action items.

- [**writing**] Section 1 (Introduction) and Section 2 (Methodology) use the acronym 'RPBH' (Randomized Permuted Block-Hadamard) multiple times before it is explicitly defined. The term appears in the abstract and intro without expansion. Define it at first use: 'randomized permuted block-Hadamard (RPBH) rotation'.
- [**writing**] Section 2.3 (RPBH) introduces the symbol `h` in Equation 5 and the text ('where h is the largest power of two...') without defining it in the notation section or immediately preceding the equation. Define `h` as the block size parameter where it first appears in the text or equation context.
- [**writing**] Section 3.1 (Setup) and Table 1 use the notation 'W4A4', 'W2A4', etc., without defining the convention. While common in quantization, an adjacent-field reader may not know 'W' stands for weight bits and 'A' for activation bits. Add a brief gloss at first use: 'W4A4 (4-bit weights, 4-bit activations)'.
- [**writing**] Section 2.3 mentions 'Rademacher sign diagonal' and 'Walsh–Hadamard matrix' without a one-sentence operational definition for a reader outside randomized linear algebra. Briefly clarify: 'Rademacher signs (random ±1 values)' and 'Walsh–Hadamard matrix (a structured orthogonal matrix with ±1 entries)'.
- [**writing**] Section 4 (Ablations) and Table 3 refer to 'Full RHT' and 'Block-RHT' without defining these acronyms. Define them at first use: 'Full Randomized Hadamard Transform (RHT)' and 'Block-Randomized Hadamard Transform (Block-RHT)'.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's logical structure is generally sound, with the core argument (rotation + distributional codebook enables calibration-free quantization) following coherently from the premises. However, there are two specific inconsistencies between the experimental setup description and the reported results/figures that require clarification.

First, there is a mismatch in the model definitions. Section 3.1 ("Setup") explicitly lists the evaluated video models as "Wan 2.1-1.3B" and "CogVideoX-2B," providing their specific frame counts and resolutions. However, Section 3.3 ("Qualitative Comparison") and the caption for Figure 3 state: "For Wan 14B video at W4A4, OrbitQuant preserves scene layout..." The "Wan 14B" model is not defined in the Setup section, nor are its experimental parameters (frames, resolution, steps) provided. While the supplementary material (Table `tab:vbench-wan14b`) does report results for Wan 14B, the main text's Setup section fails to include it in the list of evaluated models, creating a disconnect between the defined scope and the reported qualitative results.

The authors should either add Wan 14B to the Setup section with its parameters or correct the text/figure caption to refer to the evaluated Wan 2.1-1.3B model if that was the intended subject.

Second, in Section 4.1 ("Comparison between Rotation Matrix"), the text claims that RPBH is "no slower than the Full RHT." Table 1 reports RPBH latency as 0.451s and Full RHT as 0.452s. While the difference is negligible (0.001s), the phrasing "no slower" suggests RPBH is faster or equal, whereas the data shows it is technically 0.001s slower. While this is a very minor point, precise comparative language should align strictly with the tabulated data to avoid any implication of superiority where none exists.

These issues are primarily matters of textual consistency and precision rather than fundamental logical flaws in the argument. The core reasoning remains valid.

Action items.

- [**writing**] Section 3.1 (Setup) states Wan 2.1-1.3B uses '81 frames', but Section 3.3 (Video generation) and Table 2 caption refer to 'Wan 14B' for the qualitative video comparison in Figure 3. The text in Section 3.3 claims 'Wan 14B video at W4A4' is shown, but the Setup section only defines parameters for Wan 2.1-1.3B and CogVideoX-2B. Clarify if Wan 14B results were generated and if so, define its parameters in Section 3.1, or correct the figure caption/text to match the evaluated models.
- [**writing**] Section 4.1 (Ablations) text claims 'RPBH adds 0.070 s over Block-RHT' (0.451s vs 0.381s in Table 1), but the text also states 'RPBH is no slower than the Full RHT' (0.451s vs 0.452s). While numerically close, the phrasing 'no slower' implies equality or superiority, whereas the table shows RPBH is 0.001s slower. This is a minor precision issue in the comparative claim relative to the table data.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the universality and state-of-the-art status of OrbitQuant that slightly exceed the scope of the provided experimental evidence.

First, the Abstract and Conclusion assert that OrbitQuant "sets the state of the art" and is "the only method" that produces usable images at W2A4. While Table 1 confirms that OrbitQuant outperforms the *tested* calibration-based baselines (SVDQuant, ViDiT-Q, etc.) which collapse to near-zero scores at W2A4, the claim of "state of the art" is slightly overstated when looking at W4A4 results. For instance, on FLUX.1-schnell W4A4, AdaTSQ achieves an Overall score of 0.680, while OrbitQuant scores 0.703. While OrbitQuant leads, the margin is not overwhelming, and on specific sub-tasks (e.g., "Two object" on FLUX.1-schnell), AdaTSQ (0.894) actually outperforms OrbitQuant (0.881). The rhetoric implies a blanket superiority that the data shows is conditional on the specific metric and model. The claim should be narrowed to “sets the state of the art among calibration-free methods” or “achieves the best overall GenEval score across the tested models and bit-widths.”

Second, the Abstract claims the method “transfers from image to video with no per-modality tuning.” The experiments in Section 3.2 and Table 2 demonstrate success on video models (Wan, CogVideoX), but the paper does not explicitly clarify if the random permutation seeds for the RPBH rotation were shared between image and video models or re-sampled. If the permutation is re-sampled for the video domain, this constitutes a form of per-modality configuration (even if not “calibration” in the traditional sense). To support the “no tuning” claim rigorously, the authors should confirm that the exact same random seeds/parameters were applied across modalities or clarify that “no tuning” strictly refers to the absence of data-dependent calibration.

Finally, the Conclusion states the method “supports usable 2-bit weights.” While OrbitQuant significantly outperforms collapsed baselines at W2A4, the performance on Z-Image-Turbo drops from 0.754 (FP16) to 0.319 (W2A4). A >50% drop in GenEval score challenges the descriptor “usable” without qualification. The conclusion should either define “usable” (e.g., “retains non-trivial performance”) or explicitly acknowledge the degradation on Z-Image-Turbo to avoid implying that W2A4 is equally robust across all architectures.

These are primarily rhetorical overreaches that can be fixed by tightening the language in the Abstract and Conclusion to match the specific boundaries of the experimental results.

Action items.

- [**writing**] Abstract/Conclusion claim ‘sets the state of the art’ and ‘only method’ at W2A4. Table 1 shows OrbitQuant leads overall but AdaTSQ beats it on ‘Two object’ (FLUX-schnell W4A4). ‘Only method’ is true for tested baselines, but ‘SOTA’ implies universal dominance. Narrow to ‘best among calibration-free methods’ or ‘best overall score on tested models’.
- [**writing**] Abstract claims ‘no per-modality tuning’ for image-to-video transfer. Section 3.2 shows video results but doesn’t confirm if RPBH permutation seeds were shared or re-sampled. If re-sampled, this is per-modality config. Clarify if exact same seeds/params were used across modalities to support the claim.
- [**writing**] Conclusion states ‘supports usable 2-bit weights’. Table 1 shows Z-Image-Turbo drops from 0.754 (FP16) to 0.319 (W2A4), a >50% drop. ‘Usable’ is subjective here. Qualify as ‘retains non-trivial performance’ or acknowledge Z-Image-Turbo degradation to avoid overgeneralizing W2A4 robustness.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a post-training quantization (PTQ) method for Diffusion Transformers

(DiTs) that relies on mathematical rotations and pre-computed codebooks. The work does not involve human subjects, personal data collection, or the generation of sensitive content (e.g., biological agents, cyber-attack payloads, or non-consensual deepfakes). The datasets used for evaluation (GenEval, VBench) are standard public benchmarks, and the models evaluated (FLUX, Wan, CogVideoX) are existing public checkpoints.

There are no foreseeable, non-trivial risks of harm specific to this methodology that are unaddressed. The "dual-use" potential of making generative models more efficient is a generic characteristic of the field and does not constitute a specific risk requiring mitigation in this context, as the method does not lower the barrier to generating harmful content beyond what the base models already allow. The paper does not release any new training data that could violate licenses or contain PII, nor does it disclose operational vulnerabilities in live systems. Consequently, no specific safety disclosures or mitigations are missing. The verdict is accept.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling method for data-agnostic quantization, but the experimental design in the main results tables (Table 1 and Table 2) relies on single-point estimates for both the proposed method and the baselines, which is insufficient to support the strong "state-of-the-art" claims made, particularly at the margins where performance differences are small.

While the supplementary material (Table S4) provides seed robustness for OrbitQuant, the baselines (SVDQuant, AdaTSQ, ViDiT-Q) are presented as single scalar values. In the GenEval results (Table 1), OrbitQuant beats the strongest baseline (AdaTSQ) by a narrow margin (e.g., 0.703 vs 0.688 on FLUX.1-schnell W4A4). The supplementary table reports a standard deviation of 0.012 for OrbitQuant on this specific setting. The observed improvement (0.015) is comparable to the inherent noise of the evaluation (1.25x the standard deviation). Without reporting the variance for the baselines or performing a paired statistical test across seeds, it is impossible to rule out that the reported "SOTA" status is a result of a lucky random seed for OrbitQuant or an unlucky seed for the baselines.

Furthermore, the text states that baseline numbers are "taken primarily from" other papers. This introduces a significant confound: the baselines may have been evaluated with different random seeds, prompt sets, or generation parameters (e.g., guidance scale, sampler steps) than those used for OrbitQuant. If the baselines were not re-evaluated under the exact same experimental conditions (same seeds, same prompts), the comparison is not apples-to-apples. The claim that OrbitQuant is superior could simply reflect a more favorable evaluation setup for the proposed method rather than a genuine algorithmic advantage.

To close these gaps, the authors should:

1. Re-run all baseline methods (SVDQuant, AdaTSQ, etc.) using the exact same prompt

set, random seeds, and generation parameters as OrbitQuant, reporting mean ± standard deviation for all methods.

2. If re-running is not feasible, explicitly disclose the specific seeds and evaluation settings used for the cited baseline numbers and demonstrate that the variance in those settings is negligible compared to the reported performance gap.

3. For the specific cases where OrbitQuant's lead is within the reported standard deviation (e.g., FLUX.1-schnell W4A4), either report a paired statistical test (e.g., t-test or Wilcoxon signed-rank) across seeds to confirm significance, or temper the claim to "competitive with" rather than "state-of-the-art" if the difference is not statistically significant.

The qualitative results and the ablation on the rotation mechanism (Table 3) are well-supported, but the quantitative claims of superiority over existing methods require this additional statistical rigor to be convincing.

Action items.

- [**science**] Table 1 (GenEval) and Table 2 (VBench) report single-run results for OrbitQuant against baselines. While Table S4 (Suppl) shows seed variance for OrbitQuant, the baselines (SVDQuant, AdaTSQ, etc.) are presented as single numbers without variance. To rule out that OrbitQuant's lead is a lucky seed, report mean ± SD for all baselines over the same 3 seeds, or explicitly state if baseline numbers are taken from fixed literature values that preclude re-running.

- [**science**] Table 1 (GenEval) shows OrbitQuant beating AdaTSQ by 0.015 on FLUX.1-schnell W4A4 (0.703 vs 0.688). Table S4 reports a standard deviation of 0.012 for OrbitQuant on this setting. The reported gain is smaller than the run-to-run variance, making the 'state-of-the-art' claim statistically indistinguishable from noise. Report the full distribution of differences across seeds or a paired statistical test to confirm the gain is real.

- [**science**] Section 3.1 states baseline numbers are 'taken primarily from AdaTSQ... and QVGen'. If baselines are not re-run by the authors, the comparison is confounded by different evaluation seeds, prompts, or generation settings used in the original papers. To establish a fair comparison, re-run all baselines with the exact same prompt set, seeds, and generation parameters used for OrbitQuant, or disclose the specific seed/settings of the cited numbers to verify comparability.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the main results tables (Table 1 and Table 2) lacks necessary uncertainty quantification. While the paper makes strong comparative claims, the primary tables present single-point estimates for accuracy scores without accompanying standard deviations, confidence intervals, or ranges.

Specifically, in Table 1 (GenEval), OrbitQuant's Overall score of 0.703 (W4A4, FLUX.1-schnell) is presented alongside AdaTSQ's 0.680. Without error bars or seed variance, a reader cannot determine if this 0.023 difference is statistically significant or merely noise from stochasticity in the generation process or the random rotation initialization. The supplementary material (Table 4) does provide seed robustness for OrbitQuant, but this data is absent from the main comparison tables where the baselines are listed. For a fair comparison, all methods in the main tables should report mean $\pm$ standard deviation over at least 3 seeds, or the text must explicitly clarify that single-run results are shown and that variance is unquantified for the baselines.

Similarly, the ablation study in Section 4.1 and Table 3 asserts that RPBH is "the strongest" at lower bit-widths and that rotations are "within noise" at W4A4. These qualitative assessments lack quantitative backing. No standard deviations are reported for the ablation results, and no hypothesis tests are cited to confirm that the observed differences (e.g., 0.690 vs 0.696 for Haar at W4A4) are not due to random fluctuation. Given the sensitivity of diffusion generation to seeds, claiming one rotation is "strongest" based on point estimates is statistically weak. The authors should either report the standard deviation across seeds for the ablation table or conduct a paired statistical test to validate the significance of the performance gaps.

These are reporting gaps rather than fundamental flaws in the experimental design, as the raw data (per-seed results) likely exists given the supplementary table. Correcting this requires updating the tables and text to include uncertainty measures, which is a writing-level fix.

Action items.

- **[writing]** Tables 1 and 2 report single-point accuracy scores without uncertainty measures (SD/CI). While Table 4 shows OrbitQuant seed variance, baselines lack this context. Report mean $\pm$ SD over 3 seeds for all methods in main tables, or explicitly state single-run results and add a variance caveat.
- **[writing]** Table 3 claims RPBH is 'strongest' at low bit-widths based on point estimates without reported SDs or significance tests. Given generation stochasticity, this claim is unsupported. Report SD across seeds for ablation results or perform a paired test to validate the 'strongest' assertion.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the technical narrative is logical. However, there are specific instances where sentence construction and terminology consistency impede the reader's ability to parse the text on the first pass.

In the Introduction, the third paragraph contains a long, complex sentence comparing LLM and DiT activation statistics. The clause structure forces the reader to hold multiple conditions in memory before reaching the main verb. Splitting this into two shorter sentences would significantly improve flow.

A more significant issue is the inconsistent terminology regarding the rotation method. Section 2.3 defines the method as "Randomized permuted block-Hadamard (RPBH)" and uses this acronym throughout the method description. However, in the Ablation section (Section 4.1) and the discussion of the proposition, the text frequently switches to "SRHT" (Subsampled Randomized Hadamard Transform) without explicitly stating that RPBH *is* the specific SRHT implementation being used. While an expert might infer the connection, a reader moving through the paper for the first time may pause to wonder if these are two different rotation strategies being compared. The authors should unify the terminology or add a clarifying phrase (e.g., "our RPBH rotation, a variant of SRHT...") to ensure the reader knows these terms refer to the same component.

Additionally, the experimental setup section lists numerous hyperparameters (frames, resolution, steps, CFG) in dense, run-on sentences. Breaking these into distinct sentences or a structured list would make the setup easier to scan and verify.

These are primarily issues of clarity and consistency rather than fundamental structural flaws. With these targeted edits, the prose will allow the reader to move through the argument without friction.

Action items.

- [**writing**] The paper is generally well-structured and the technical narrative is logical. However, there are specific instances where sentence construction and terminology consistency impede the reader's ability to parse the text on the first pass. In the Introduction, the third paragraph contains a long, complex sentence comparing LLM and DiT activation statistics. The clause structure forces the reader to hold multiple conditions in memory before reaching the main verb. Splitting this into two shorter se